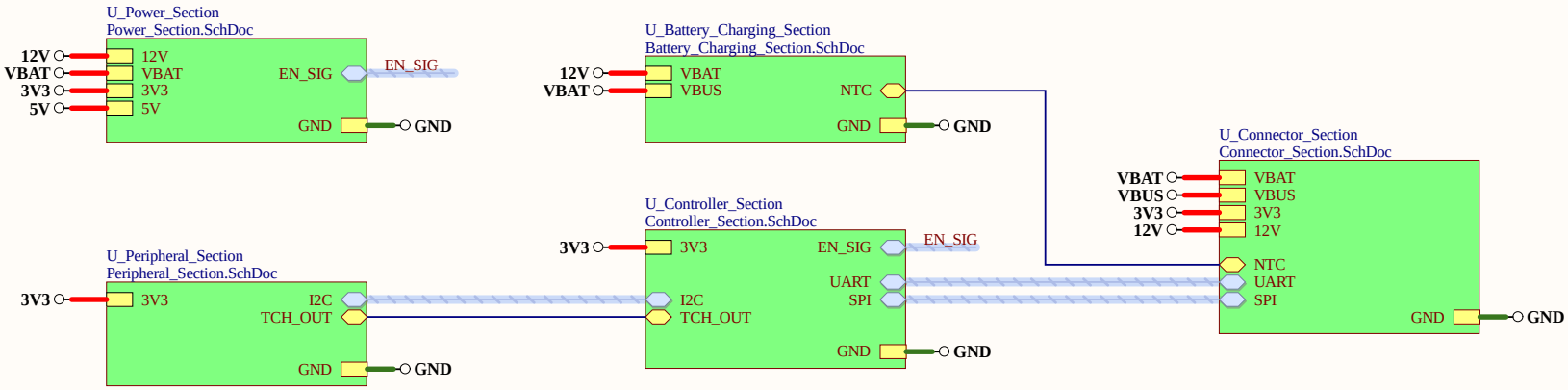
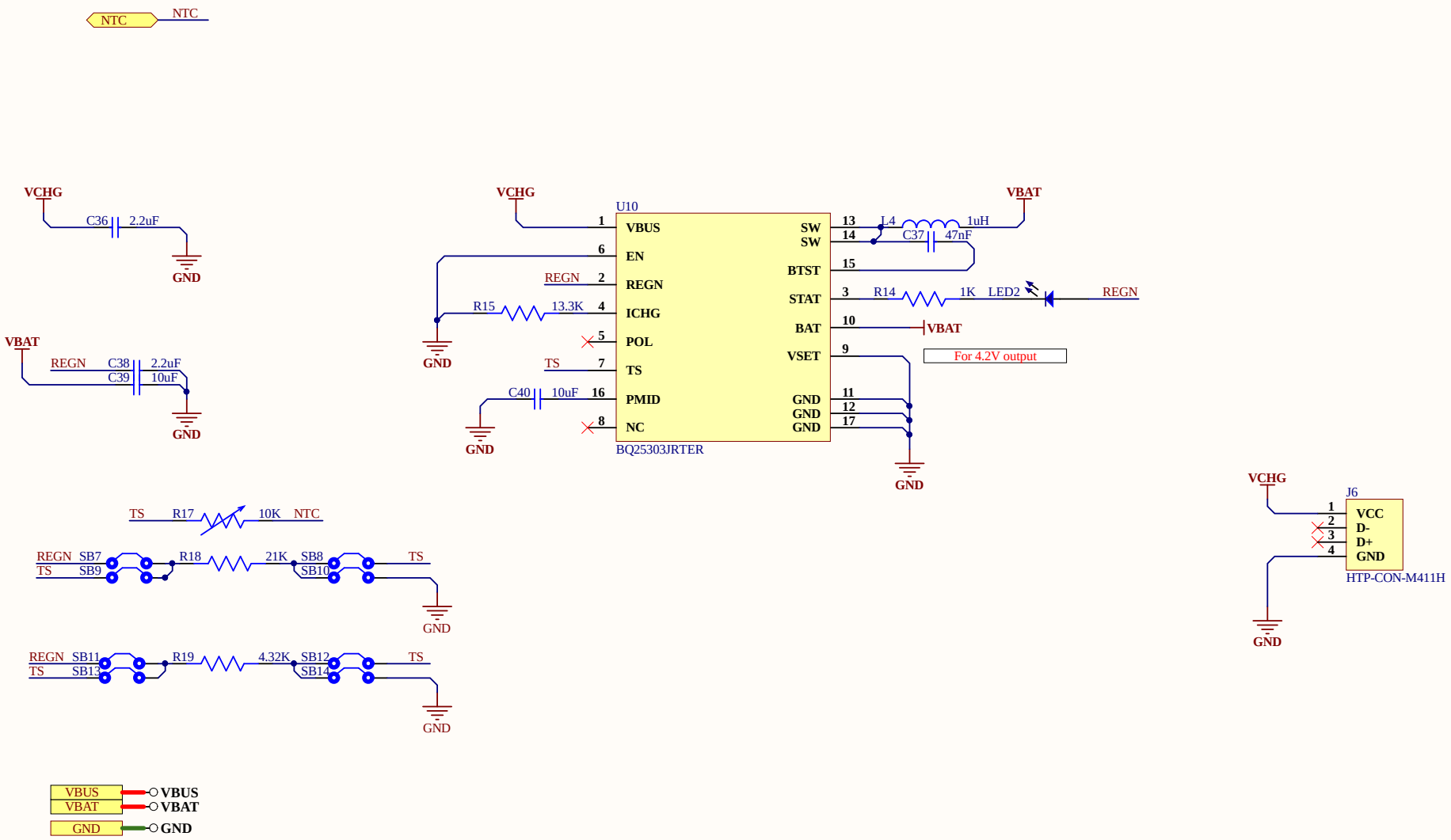
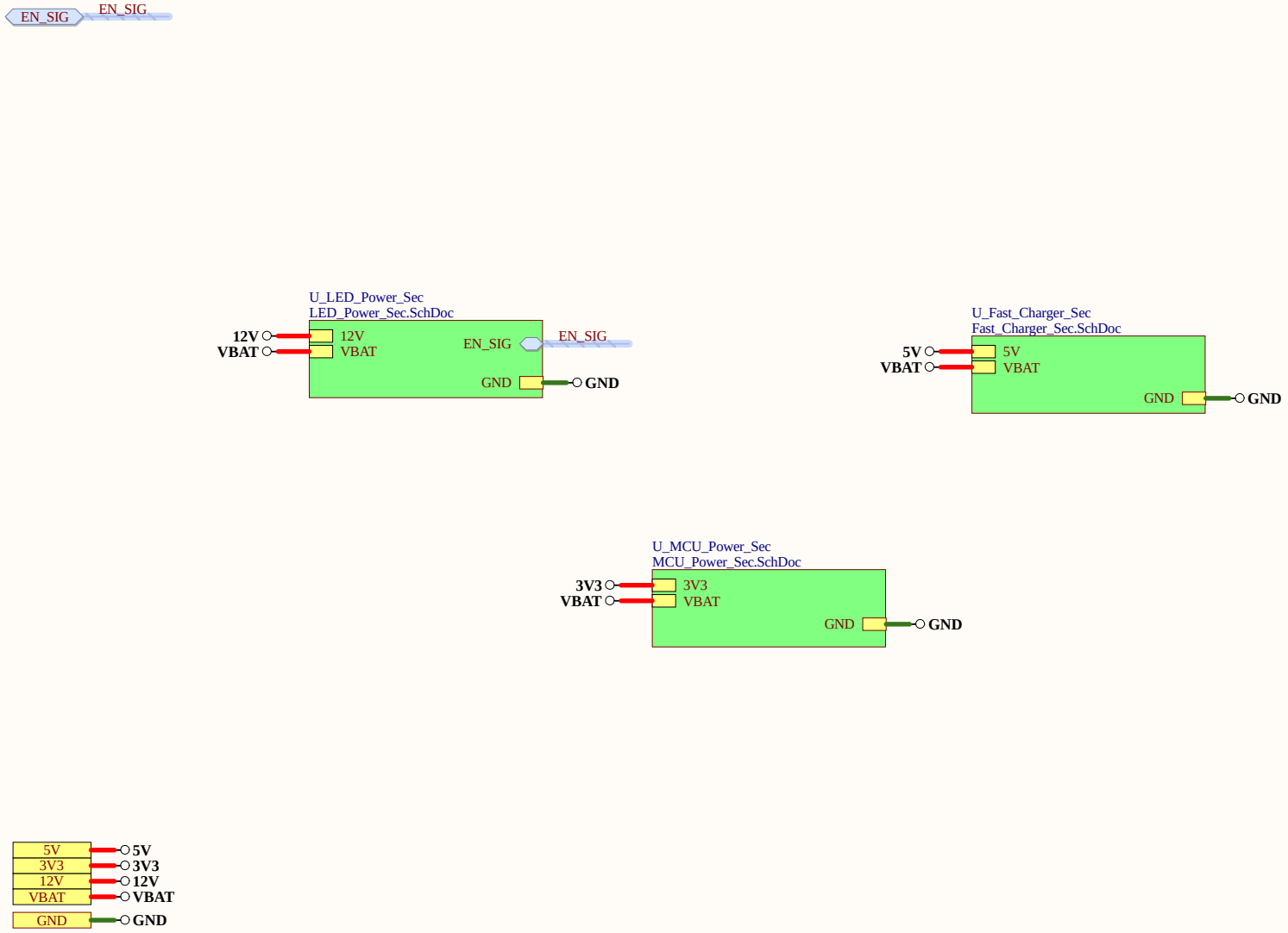
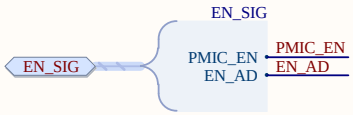
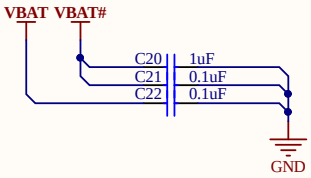
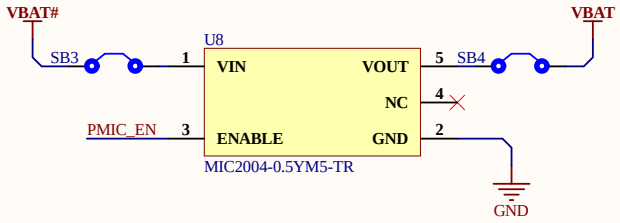

U_Power_Section
Power_Section.SchDoc
12V
VBAT
3V3
5V
12V
VBAT
3V3
5V
EN_SIG
EN_SIG
GND
GND
U_Battery_Charging_Section
Battery_Charging_Section.SchDoc
12V
VBAT
VBAT
VBUS
NTC
GND
GND
U_Connector_Section
Connector_Section.SchDoc
VBAT
VBUS
3V3
12V
VBAT
VBUS
3V3
12V
NTC
UART
SPI
GND
GND
U_Controller_Section
Controller_Section.SchDoc
3V3
3V3
EN_SIG
EN_SIG
UART
SPI
I2C
TCH_OUT
I2C
TCH_OUT
GND
GND
U_Peripheral_Section
Peripheral_Section.SchDoc
3V3
3V3
I2C
TCH_OUT
GND
GND

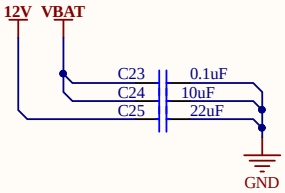

NTC
NTC
VCHG
C36
2.2uF
GND
VBAT
REGN
C38
2.2uF
C39
10uF
GND
VCHG
U10
1
VBUS
6
EN
REGN
2
REGN
R15
13.3K
4
ICHG
5
POL
TS
7
TS
C40
10uF
16
PMID
8
NC
GND
BQ25303JRTER
SW
13
SW
14
BTST
15
STAT
3
BAT
10
VSET
9
GND
11
GND
12
GND
17
L4
1uH
C37
47nF
VBAT
R14
1K
LED2
REGN
VBAT
For 4.2V output
GND
TS
R17
10K
NTC
REGN
SB7
TS
SB9
R18
21K
SB8
SB10
TS
GND
REGN
SB11
TS
SB13
R19
4.32K
SB12
SB14
TS
GND
VCHG
J6
1
2
3
4
VCC
D-
D+
GND
HTP-CON-M411H
GND
VBUS
VBAT
GND
VBUS
VBAT
GND

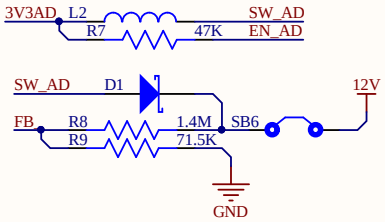

EN_SIG
EN_SIG
U_LED_Power_Sec
LED_Power_Sec.SchDoc
12V
VBAT
EN_SIG
GND
12V
VBAT
EN_SIG
GND
U_Fast_Charger_Sec
Fast_Charger_Sec.SchDoc
5V
VBAT
5V
VBAT
GND
GND
U_MCU_Power_Sec
MCU_Power_Sec.SchDoc
3V3
VBAT
3V3
VBAT
GND
GND
5V
3V3
12V
VBAT
GND
5V
3V3
12V
VBAT
GND

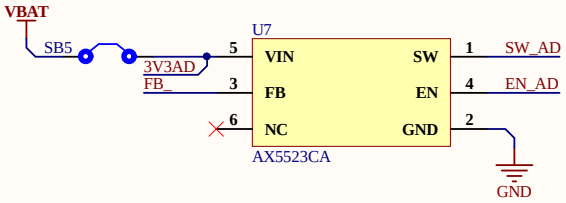

EN_SIG
EN_SIG
PMIC_EN
EN_AD
PMIC_EN
EN_AD

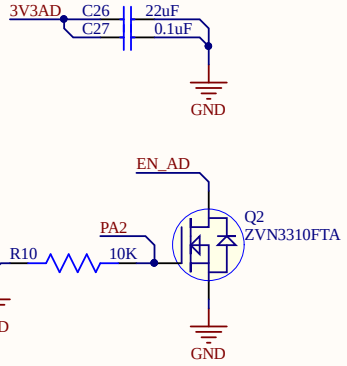

VBAT
VBAT#
C20
C21
C22
1uF
0.1uF
0.1uF
GND

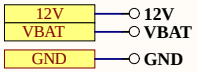

VBAT#
SB3
1
VIN
VOUT
5
SB4
VBAT
NC
4
PMIC_EN
3
ENABLE
GND
2
U8
MIC2004-0.5YM5-TR
GND

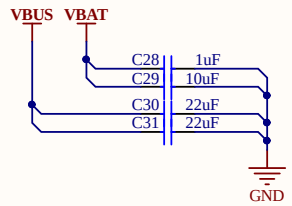

12V
VBAT
C23
C24
C25
0.1uF
10uF
22uF
GND

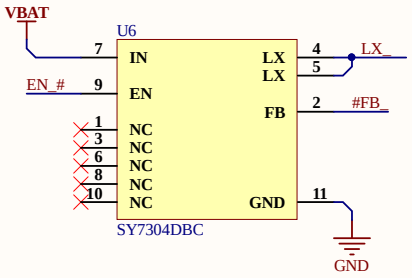

3V3AD
L2
SW_AD
R7
47K
EN_AD
SW_AD
D1
12V
FB
R8
1.4M
SB6
R9
71.5K
GND

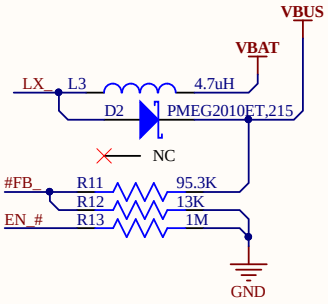

VBAT
SB5
U7
5
VIN
SW
1
SW_AD
3V3AD
FB_
3
FB
EN
4
EN_AD
6
NC
GND
2
AX5523CA
GND

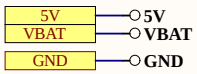

3V3AD
C26
C27
22uF
0.1uF
GND
EN_AD
PA2
Q2
ZVN3310FTA
R10
10K
GND
GND

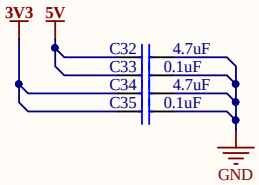

12V
VBAT
GND
12V
VBAT
GND

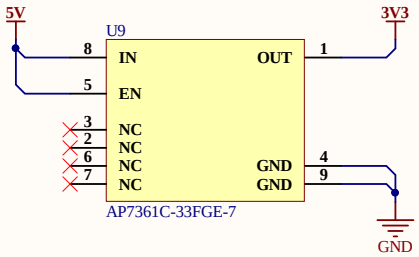

VBUS
VBAT
C28
1uF
C29
10uF
C30
22uF
C31
22uF
GND

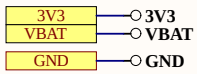

VBAT
U6
7
IN
EN_#
9
EN
1
NC
3
NC
6
NC
8
NC
10
NC
LX
4
LX
5
FB
2
#FB_
GND
11
SY7304DBC
LX_
GND

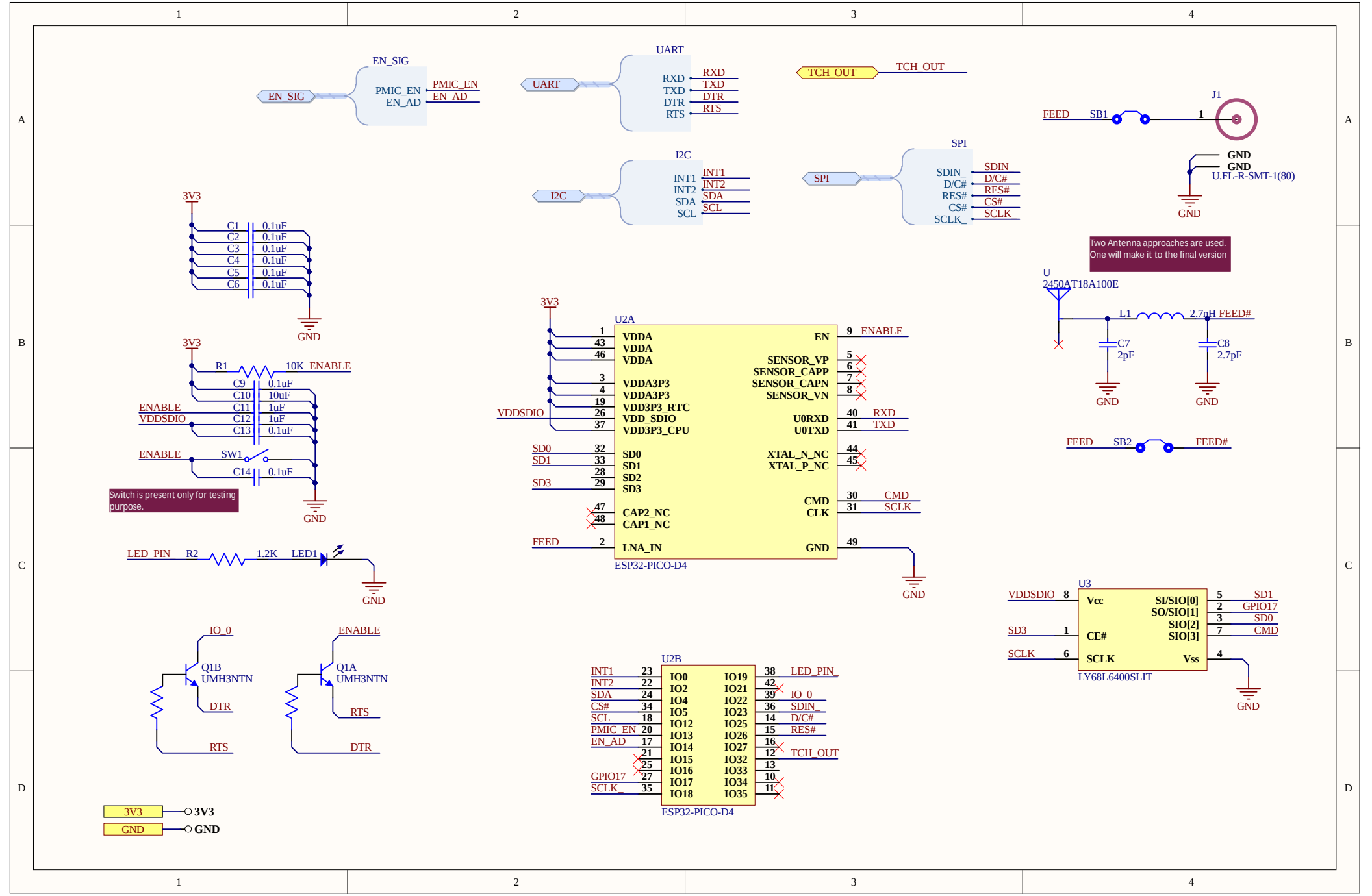

VBUS
VBAT
LX_
L3
4.7uH
D2
PMEG2010ET,215
NC
#FB_
R11
95.3K
R12
13K
EN_#
R13
1M
GND

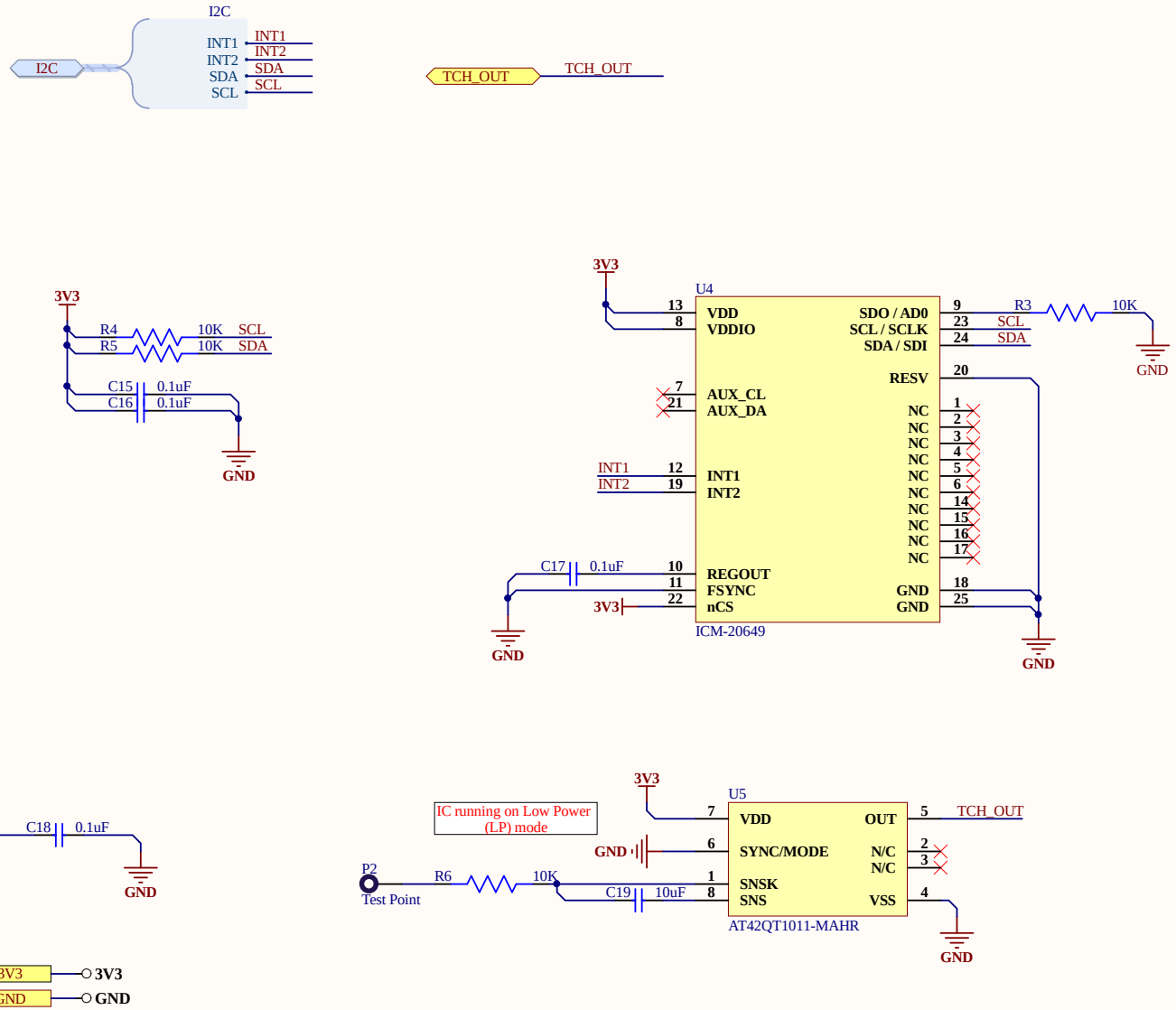

5V
VBAT
GND
5V
VBAT
GND

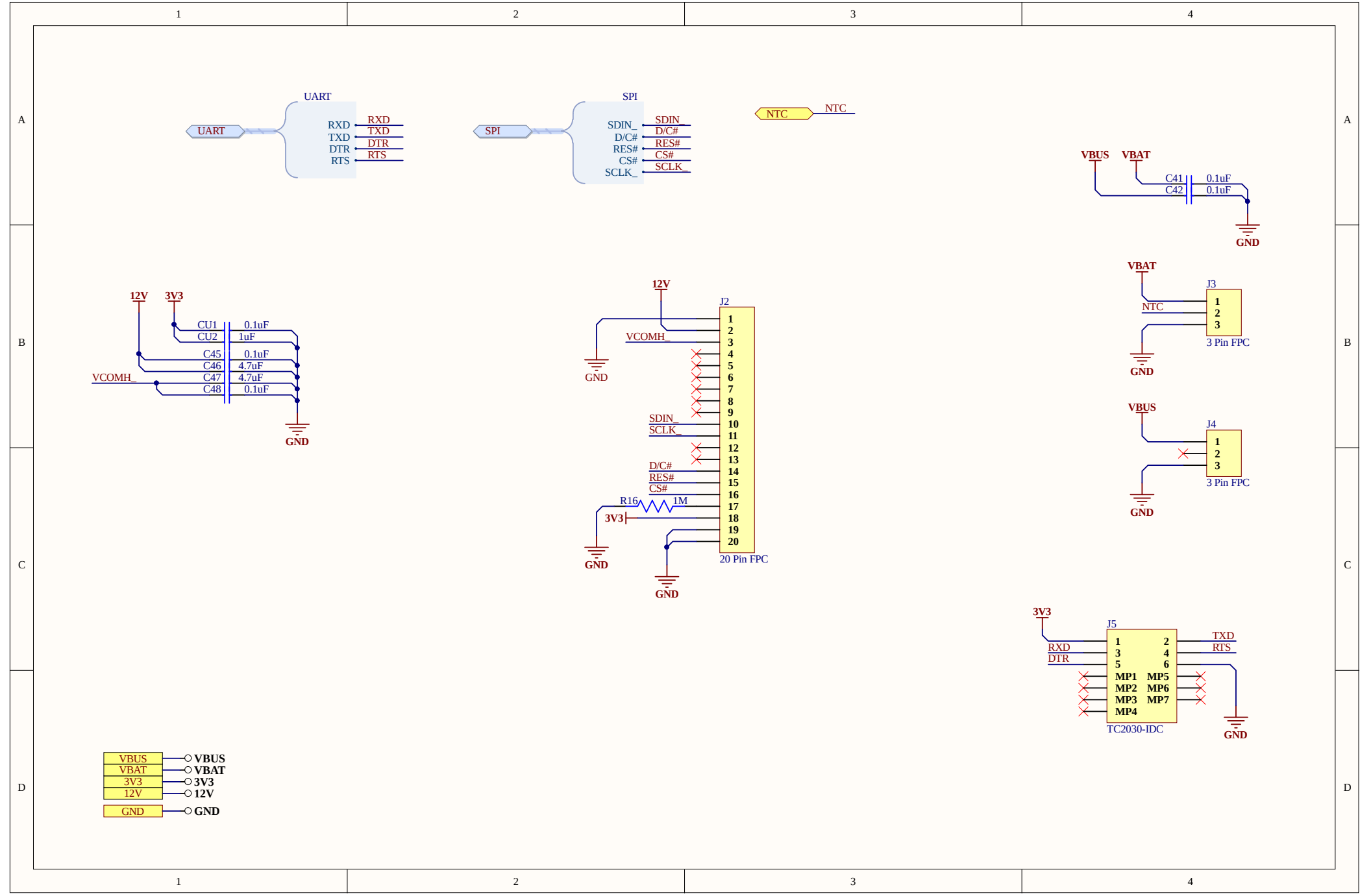

3V3
5V
C32
4.7uF
C33
0.1uF
C34
4.7uF
C35
0.1uF
GND


5V
3V3
U9
8
IN
OUT
1
5
EN
3
NC
2
NC
6
NC
GND
4
7
NC
GND
9
AP7361C-33FGE-7
GND


3V3
VBAT
GND
3V3
VBAT
GND


EN_SIG
PMIC_EN
EN_AD
UART
RXD
TXD
DTR
RTS
TCH_OUT
I2C
INT1
INT2
SDA
SCL
SPI
SDIN_
D/C#
RES#
CS#
SCLK_
J1
FEED
SB1
GND
U.FL-R-SMT-1(80)
3V3
C1 0.1uF
C2 0.1uF
C3 0.1uF
C4 0.1uF
C5 0.1uF
C6 0.1uF
Two Antenna approaches are used.
One will make it to the final version
U
2450AT18A100E
L1 2.7nH FEED#
C7 2pF
C8 2.7pF
R1 10K ENABLE
C9 0.1uF
C10 10uF
C11 1uF
C12 1uF
C13 0.1uF
VDDSDIO
SW1
C14 0.1uF
Switch is present only for testing purpose.
SB2
FEED#
U2A
VDDA
VDDA3P3
VDD3P3_RTC
VDD_SDIO
VDD3P3_CPU
EN
SENSOR_VP
SENSOR_CAPP
SENSOR_CAPN
SENSOR_VN
U0RXD
U0TXD
XTAL_N_NC
XTAL_P_NC
SD0
SD1
SD2
SD3
CMD
CLK
CAP2_NC
CAP1_NC
LNA_IN
ESP32-PICO-D4
LED_PIN_ R2 1.2K LED1
IO_0
ENABLE
Q1B UMH3NTN
Q1A UMH3NTN
DTR
RTS
U2B
IO0
IO2
IO4
IO5
IO12
IO13
IO14
IO15
IO16
IO17
IO18
IO19
IO21
IO22
IO23
IO25
IO26
IO27
IO32
IO33
IO34
IO35
U3
Vcc
CE#
SCLK
SI/SIO[0]
SO/SIO[1]
SIO[2]
SIO[3]
Vss
LY68L6400SLIT
3V3
GND


I2C
INT1
INT2
SDA
SCL
TCH_OUT
3V3
R4
R5
10K
C15
C16
0.1uF
GND
U4
VDD
VDDIO
AUX_CL
AUX_DA
INT1
INT2
REGOUT
FSYNC
nCS
SDO / AD0
SCL / SCLK
SDA / SDI
RESV
NC
R3
ICM-20649
C17
C18
IC running on Low Power (LP) mode
P2
Test Point
R6
C19
10uF
U5
VDD
SYNC/MODE
SNSK
SNS
OUT
N/C
VSS
AT42QT1011-MAHR


1
2
3
4
A
B
C
D
UART
UART
RXD
TXD
DTR
RTS
RXD
TXD
DTR
RTS
SPI
SPI
SDIN_
D/C#
RES#
CS#
SCLK_
SDIN_
D/C#
RES#
CS#
SCLK_
NTC
NTC
VBUS
VBAT
C41
C42
0.1uF
0.1uF
GND
12V
3V3
CU1
CU2
0.1uF
1uF
C45
C46
C47
C48
0.1uF
4.7uF
4.7uF
0.1uF
VCOMH_
GND
12V
J2
VCOMH_
GND
SDIN_
SCLK_
D/C#
RES#
CS#
R16
1M
3V3
GND
20 Pin FPC
VBAT
J3
NTC
GND
3 Pin FPC
VBUS
J4
GND
3 Pin FPC
3V3
J5
RXD
DTR
TXD
RTS
MP1
MP2
MP3
MP4
MP5
MP6
MP7
GND
TC2030-IDC
VBUS
VBAT
3V3
12V
GND
VBUS
VBAT
3V3
12V
GND